Enhancing Enhancer-Promoter Interaction Prediction via Gene Regulatory Inference and Hybrid Deep Learning Architectures

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Abstract

Enhancer-promoter interactions (EPI) control gene regulations through the distal communication with the target promoters, enabling it through chromatin looping. Understanding the EPI prediction correctly is highly important for learning and understanding the disease mechanisms, and to infer how they work with certain Non coding DNA segments. However Current methods face 3 limitations, namely the relational gap between the advanced position-aware sequence encoding and simplistic input representation in on going inference pipelines, not enough fusion of sequential, (both positive and negative) and epigenic features, and limited capabilities of interaction models to infer properly within time or computational efficiency.

We propose a Hybrid model consisting of POCD-KansformerEPI, a deep learning framework that combines Position Specific Nucleotide Density-based encoding (POCD-ND) with a hybrid Kansformer architecture for better EPI Predictions. We employ POCD-ND encoding instead of usual sparse One hot Encoding and motif clustering pattern through Convolutional Neural Netowrks(CNNs), making sure that we ignore the spacial and mechanistic feature collection loss for K-mer($2 \leq k \leq 5$) counting. The processed data gets sequential contents by bidirectional Long Short Term Memory(LSTM) and pipelined through the Kolmogorov-Arnold Networks of The Kansfromer with parameterized activations functions to model complex non-Linear regulatory relations. The Semi-Multimodal learning concurrently enhances the interaction classification and also learns better distant gene classification and EPI is optimised through distant proximal policies. Meanwhile Class Activation Maps provide interpretability by analysing predictive sequence features.

The framework is tested and analysed across multiple cell lines (GM12878, HeLa-S3, HMEC, IMR90, K562, NHEK) using distance-matched negative sampling to prevent spurious biases. Our model is expected to achieve superior cross-cell-line generalization while revealing biologically precious EPI regulation patterns, enabling close testable hypotheses about disease-associated non-coding variants.

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Publication List

[Optional] The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

1.

Conference Papers

1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

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Chapter 1

Introduction

A pivotal change has been witnessed in the gene regulation for the past twenty years. It changed the view of scientists to understand biological deposition. In early decades, scientists first decoded the human genome and found coding sequences as the key genetic material. On the other hand, non-coding sequences, which were the majority part, around ninety-eight to ninety-nine percent, were junk DNA. But later they discovered that this portion plays an essential role in gene regulation. The dominating non-coding DNA that determines gene expression are called enhancers and promoters. Enhancers are defined as cis-regulatory sequences that work from a distance but control the process. Whereas promoters are defined as proximal cis-regulatory sequences that act as starting positions for the cellular machinery that reads genetic information. Despite being physically separated in the linear DNA sequence, enhancers boost the process dramatically. Because of DNA folds and loops with the help of proteins. This method is known as topologically associating domains (TADs). In the genome, regulatory elements interact in this complex way.

In this internal intricateness, in modern biology it is challenging to find which enhancers regulate which genes. This interaction is known as enhancer-promoter interaction (EPI). Its prediction is the intersection of computational genomics and molecular biology. This prediction helps to understand how genetic variations contribute to diseases such as cancer or developmental disorders.

1.1 Project Overview

This project introduces a modern deep learning techniques to predict enhancer promoters. interactions by combining advanced sequence encoding methods. Many existing models converts DNA sequence for simplicity or sacrifices precise positions which carry crucial biological information. Our proposed solution, POCD-KansformerEPI, is designed to cover those limitations. Firstly, the method Position Specific Nucleotide Density-based encoding (POCD-ND) [1] to capture nucleotide location and cluster by sequence. Secondly, a hybrid architecture combining Kolmogorov-Arnold Networks (KANs) with Transformers [15] for the long-range dependencies. Also using special information, our solution represents regulatory factors in non-additive ways. With our introduced model, significantly better predictions are possible.

1.2 Motivation

The way current methods represent DNA sequences for computational analysis is a major concern.

1.2.1 The Representation Bottleneck: Beyond Linear Sequences

Proteins known as transcription factors bind to particular short sequences known as motifs to activate regulatory DNA. However, the presence of the correct motif sequence is not the only factor that determines whether a transcription factor binds; other factors include how closely these motifs cluster together and how far apart they are from other binding sites. This complexity presents a challenge to conventional encoding schemes. Position tracking is maintained by one-hot encoding, but it produces very sparse, high-dimensional matrices that are computationally costly and prone to overfitting when training data is scarce. Conversely, basic k-mer counting completely loses track of motif locations while compressing sequences into compact vectors. Even though these sequences may have very different biological activities, using this method, a sequence with binding sites dispersed throughout and one with the same sites clustered in the center appear identical. To capture nucleotide distribution at different positions and recognise conserved patterns we adopt POCD-ND encoding method that distinguish functional enhancers from inactive sequences [1].

1.2.2 Modelling Nonlinearity in Gene Regulation

Gene regulation is another important thing because it is uncertain.Chromatin states contradict in complex ways When an enhancer activates a gene . Transcription factors frequently bind together, and regulatory proteins can form phase-separated moisture that concentrates the transcriptional machinery. The most common regulatory pattern occurs when a single promoter is joined on by multiple enhancers, which results in complex interactions between inputs that dictate the final output.

ReLU is a common activation function used by most of the neural networks. But they struggle to capture dynamics. For example, Because of non-linear relationships, Multi Layer Perceptrons (MLPs) make training difficult even though it is theoretically capable. To overcome this hurdle, we used the KansformerEPI framework [15], which has trainable activation functions for neuron connections. This difference makes multiple enhancers regulate complicated non-linear functions more smoothly to their targets.

1.2.3 Capturing Long-Range Dependencies via Attention

The sheer scale of genomic distances presents another challenge. Enhancers and their target promoters can sit hundreds of thousands of base pairs apart, yet they must somehow communicate effectively. Earlier neural network designs like Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks work reasonably well for shorter sequences but tend to suffer from vanishing gradient problems when dealing with very long inputs.

The Transformer architecture offers a solution through its self-attention mechanism. Rather than processing sequences one element at a time, attention allows every part of the input to directly influence every other part, regardless of how far apart they sit in the linear sequence.

By building a Transformer encoder into our framework, we ensure that our model can learn the global, long-range patterns that characterise enhancer-promoter communication across large genomic spans.

1.2.4 The Interpretability Imperative

Finally, this research is motivated by the growing recognition that “black box” predictions, however accurate, have limited utility in biology. Knowing that an enhancer probably interacts with a gene is helpful, but understanding why that interaction occurs is far more valuable. This type of understanding can be helpful for designing the follow-up experiments. It can help to interpret disease-associated genetic variants, and also in connecting computational predictions to biological mechanisms.

Recent work has shown that interpretability and performance need not be mutually exclusive. Which parts of an input sequence can influence predictions most effectively can be highlighted by the Class Activation Maps method [10]. Proximal Policy Optimization [6] which is a reinforcement learning technique. It can identify which are the most essential features for classification. Our motivation includes merging these interpretability tools so that we can identify the “core regions” of enhancers which are the specific subsequences that drive their regulatory activity.

1.3 Objectives

To create a predictive deep learning framework which can link all the enhancers to their targeted gene by remaining generalizable and interpretable. Our objectives are:

1. **Develop a high-confidence sequence encoder:** Finding the positional patterns and motif densities of an enhancer sequence by implementing and optimising the Position Specific Nucleotide Density-based encoding (POCD-ND). This should tame the sparsity issues of One-Hot Encoding and also overcome the spatial information loss of simple k-mer counting.

2. **Assemble a Hybrid Nonlinear Architecture:** We want to design a neural network which will combine Convolutional Neural Networks (CNN) to detect the local patterns and BiLSTMs to capture sequential contexts and a Kansformer encoder which combines KAN with transformers which will model the global, nonlinear regulatory interactions.

3. **Apply multi-task learning:** We want to build a training framework that will simultaneously optimise the prediction of interactions between enhancer-promoter and also the distance regression. This double objective will be helping the model learning the spatial constraints that controls the communication between enhancer and promoter.

4. **Ensuring the model interpretability:** Combine Proximal Policy Optimization (PPO) to select features and Class Activation Maps (CAM) for visualising the predictive regions.This tools will help point out the sequence features and prediction that drives the model’s prediction.

5. **Validating over diverse cell-line:** Demonstrate that the model can work not only on specific cell types but beyond by training with the data of multiple cell lines like GM12878 and HeLa-S3 and testing on independent cell lines like HMEC, IMR90, K562, NHEK. This should confirm that the model can learn universal regulatory principals instead of working on specific cell type patterns.

1.4 Methodology

We followed a systematic approach which is an efficient pipelining that will transform the raw genomic data into reliable predictions of interactions between enhancers and promoters. This method has 5 core phases where each of them addresses the specific biological and computational challenges.

1.4.1 Phase 1: Data Acquisition, Curation, and Preprocessing

Good data is the foundation of any kind of machine learning project. In this phase we merge a vast dataset from multiple sources. Interaction data comes from the Benchmark datasets for enhancer-gene interactionsI (BENGI) repository [18], which provides the data of enhancer-promoter pairs from Hi-C and ChIA-PET assays across diverse cell types which are experimentally validated. For enhancer definitions we used some common resources like CATlas [10]and EnhancerAtlas 2.0.

Generating appropriate negative controls is a tough step. If we pair random promoters with random enhancers it will make the model learn by cheating that closer pairs are more likely to interact with each other. To avoid this issue, we used distance-matched negative sampling [15, 18]: for each of the legit interacting pair that are separated by distance d, we have selected a non interacting pair which is at a similar distance. To eliminate other potential biases we are using GC content and repeat density.

1.4.2 Phase 2: Advanced Enhancer Encoding via POCD-ND

In this phase, we show how we are representing the DNA sequences. Here, we use POCD-ND encoding to create position-aware feature matrices and avoid treating sequences as simple strings. The process works by segmenting sequences into overlapping k-mers.(we have used trinucleotides based on previous optimised results), creating position-specific frequency matrices for both the positive and the negative examples, normalising these frequencies into density values and then we calculate the propensity scores which highlights those positions where enhancers and non-enhancers differ the most. The result gives both the presence and the positional distribution of sequence motifs which is a dense feature representation.

1.4.3 Phase 3: Hybrid Feature Integration via Kansformer Architecture

Information is processed in the neural network architecture through two main parallel streams that are sequence stream and epigenetic stream. The sequence stream uses the POCD-ND encoded matrix through a 1D CNN and detects the local motif patterns and then uses a BiLSTM to capture the sequential arrangement of these motif patterns. The epigenetic stream works by processing the chromatin features through its own CNN pathway which include histone modifications, CTCF binding, and accessibility signals.

These streams merge in the Kansformer Encoder, which uses Multi-Head Self-Attention to weigh the importance of different features and KAN layers to model nonlinear interactions. Unlike standard feed-forward layers with fixed activation functions, KAN layers learn flexible activation functions that can adapt to the complex relationships characteristic of gene regulation.

1.4.4 Phase 4: Prediction with Multi-Task Learning

The model produces outputs through two prediction heads. The primary head predicts whether a given enhancer-promoter pair interacts (binary classification). An auxiliary head predicts the genomic distance between the pair (regression). Training both tasks together forces the network to learn representations that encode spatial information, which helps prevent overfitting and improves generalisation.

1.4.5 Phase 5: Interpretability, Optimisation, and Validation

To make our model interpretable, we incorporate a PPO agent that learns to select the most important features, effectively creating a feature importance ranking. We also use Class Activation Maps to visualise which parts of input sequences most strongly influence predictions. For validation, we use cross-cell-line evaluation, training on some cell types and testing on others to confirm generalisation. We also perform in silico perturbation experiments, computationally mutating predicted important regions to verify that these mutations actually change predictions as expected.

1.5 Project Outcome

Successfully completing this project should yield several valuable contributions. Most directly, we expect to produce a model that outperforms existing methods for predicting enhancer-promoter interactions, particularly in cross-cell-line generalisation. The model should capture regulatory patterns that other simple approaches will miss because our model is an integration of POCD-ND encoding with the Kansformer architecture.

The interpretability components not only provide raw performances but also with other biological insights. We can build a hypothesis on regulatory mechanisms by identifying that which sequence positions and chromatin features will drive predictions, which can be tested with experimentations. It will make the model more useful for more practical applications like which variants can cause diseases in the non-coding regions of DNA. Capturing positional information and nonlinear interactions can be very crucial in genomics. Our methodology can be used like a template to handle this type of crucial matters. To avoid common biasesThe distance-matched negative sampling strategy is a very efficient way of training the data.

1.6 Organization of the Report

The remaining part of the report contains two chapters. Chapter 2 has the project background that includes biological preliminaries on gene regulation, a literature review organised by topic, and a gap analysis of the current approaches that our project addresses. Chapter 3 is about the project design in detail which covers the requirements analysis, the complete methodology with all the mathematical formulas, project planning, and allocating tasks.

Chapter 2

Background

2.1 Preliminaries

2.1.1 The Regulatory Landscape: Enhancers and Promoters

Gene expression in complex organisms involves an intricate dance between proteins and DNA. On one side are transcription factors and other regulatory proteins that move around the cell nucleus. On the other are specific DNA sequences that these proteins recognise and bind to. Understanding this interplay is essential for understanding how cells control which genes are active and which remain silent.

Promoters occupy the region immediately before a gene begins. In order to initiate transcription for RNA Polymerase II, the enzyme that transcribes DNA into mRNA, a docking site is provided by promoters. In this sense, promoters act like on-switches for genes. However, promoters can only drive a low level of gene activity on their own.

Enhancers perform a different function. They amplify transcription, instead of initiation. Enhancers can improve the trsanscription of their target gene by orders of size once active. Their flexibility makes them quite interesting; as they can work from positions upstream or downstream of their target genes, surrounded by introns of other genes, and covering distances that can stretch hundreds of kilobases. They can also function regardless of their target gene's orientation relative.

This distance autonomy raises questions about the mechanism. How can a distant gene's transcription be influenced by a piece of DNA sitting so far away? Part of the answer can be found if we delve deep into the protein that binds with enhancers, including transcription factors that detect specific short sequence motifs, along with co-activators like p300/CBP and the large mediator complex that stretches enhancers to the core transcriptional machinery.

Tissue specificity is another hurdle. An enhancer that activates neuron gene might be completely dormant in liver cells. Due to different cell types exhibiting different transcription factor combinations, this specificity arises. An enhancer activates transcription only when the right proteins are available to bind with it.

2.1.2 Chromatin Looping and the 3D Genome

The puzzle of how distant enhancers reach their target promoters becomes clearer once we consider the three-dimensional organisation of chromosomes. DNA in our cells does not exist as straight fibres. Instead, it wraps around histone proteins to form nucleosomes, which further compact into higher-order structures. Beyond this basic packaging, the genome adopts a complex three-dimensional architecture involving loops and domains.

At the megabase scale, chromosomes partition into Topologically Associating Domains (TADs). Within a TAD, DNA sequences interact frequently with one another, but interactions across TAD boundaries are much rarer. These boundaries act as insulators, preventing enhancers in one domain from inappropriately activating genes in neighbouring domains. Several proteins help establish and maintain this organisation. CTCF recognises specific DNA sequences and often marks TAD boundaries. The cohesin complex physically extrudes DNA loops and helps anchor them at CTCF sites.

Within TADs, specific chromatin loops bring enhancers into physical proximity with their target promoters. When an enhancer and promoter are held together by such a loop, the proteins bound at the enhancer can directly interact with and stimulate the transcriptional machinery assembled at the promoter. This looping model explains how linear distance in the genome becomes less important than three-dimensional proximity.

For computational prediction, this architecture poses significant challenges. A model must somehow learn the rules governing which enhancers can reach which promoters—not just recognising the sequence features that mark enhancers and promoters, but also understanding the "syntax" that determines which elements can physically contact each other.

2.1.3 Data Modalities and Experimental Assays

Two main types of experimental data are drawn on by computational approaches in order to predict enhancer-promoter interaction.

The first group includes epigenomic features that describe the chromatin state at distinct genomic positions. ChIP-seq (Chromatin Immunoprecipitation followed by sequencing) maps where particular proteins bind throughout the genome. Enhancer prediction relevant targets include CTCF, different transcription factors, and histone mutations. Active enhancers commonly exhibit enrichment for H3K4me1 (a modification correlated with enhancer priming) and H3K27ac (a mark of active enhancers), while active promoters show H3K4me3. DNase-seq and ATAC-seq measure chromatin accessibility regions where DNA is not tightly wrapped around nucleosomes and is therefore accessible for protein binding. Open chromatin generally indicates active regulatory regions.

The second group includes interaction data that directly measures physical contacts between genomic areas. Hi-C provides a genome-wide view of three-dimensional contacts although at relatively low resolution. ChIA-PET (Chromatin Interaction Analysis by Paired-End Tag sequencing) offers higher resolution by concentrating on interactions moderated by specific proteins, such as RNA Polymerase II or CTCF. Prediction models are often developed by using these interaction datasets as ground truth labels.

Although epigenomic data is highly instructive, it comes with practical constraints. These experiments are expensive, time-consuming, and must be performed separately for each cell type of interest. A model trained to rely on H3K27ac ChIP-seq data, for instance, cannot make predictions for a new tissue type without first generating that specific dataset. This limitation motivates sequence-based prediction approaches that rely only on the DNA sequence itself, which is available for any organism with a reference genome.

2.2 Literature Review

The field of computational gene regulation has evolved rapidly over the past couple years. This review paper examines key research contributions, organised subjectively from enhancer identification to interaction prediction to biological mechanism studies.

2.2.1 The Evolution of Enhancer Identification

The field of computational regulation of gene expression has advanced rapidly over the past decade. This review examines key research contributions, organised propositionally from enhancer identification to interaction prediction to biological mechanism studies.

One of the most important contribution is the PDCNN model [1], it established Position-Specific Nucleotide Density-based encoding (POCD-ND). The authors admitted that the way nucleotides scatter themselves at different locations within a sequence is not captured by traditional sequence encoding. By computing position-aware density features that identifies enhancers from non-enhancers based on where certain motifs cluster, POCD-ND tackles this problem . The model clearly showed advantage from position-aware encoding, achieving over 95% accuracy with an MCC of 92.6%.

The ADH-Enhancer framework [3] takes a distinct strategy, which involves elements of natural language processing. The authors employed an AWD-LSTM-based language model on genomic sequences using an unsupervised technique, creating a feature space that contains the statistical patterns of regulatory DNA prior to any supervised learning implementation.For classification, this pre-trained model was then combined with a CNN and attention mechanism. This method showed how transfer learning from language modeling may benefit genomic applications by exceeding previous strategies by 7% in accuracy and 14% in MCC.

DeepEnhancerPPO [6] deals with the interpretability problem head-on. This architecture uses Proximal Policy Optimization by incorporating a reinforcement learning component with ResNet and Transformer modules. The PPO agent learns to choose which features are important for classification, by effectively filtering out noise as well as maintaining strong performance. This approach offers a link between biological interpretability that researchers need to generate testable hypotheses and high predictive accuracy.

The Enhancer-MDLF framework [9] addresses cell-type specificity by integrating multiple input modalities: DNA2Vec embeddings that capture sequence context and motif frequency features that summarise transcription factor binding site patterns. By combining these inputs through transfer learning, the model generalises well across different cell lines without requiring extensive retuning of parameters.

Enhancer Detector [10] explores whether enhancer recognition is evolutionarily conserved.

Trained on human data, this CNN-based model generalises surprisingly well to mouse and fruit fly genomes. The authors used Class Activation Maps to identify which sequence regions drive predictions and validated these through computational mutagenesis experiments. Fine-tuning on relatively small datasets allowed the model to adapt to new species, suggesting that fundamental features of “enhancerness” are shared across evolution.

Work on explainable AI approaches [25] offers a complement to deep learning methods. One study applied Type-2 Fuzzy Logic combined with evolutionary computation to predict enhancers from histone modification data. Beyond making predictions, this rule-based system generated explicit logical rules explaining the combinations of epigenetic marks that define different enhancer classes. Such transparency helps researchers understand what distinguishes active enhancers from inactive sequences or primed enhancers waiting for activation signals.

2.2.2 Evolution of Enhancer–Promoter Interaction Prediction

Predicting which enhancers control which genes presents a more challenging problem than simply recognizing the enhancers. Interaction prediction needs understanding how two potentially distant sequences connect with each other.

Dynamic Feature fusion, which is a significant architectural advance, is represented by EPI-DynFusion[11]. From different processing branches, Many previous models simply merged features. EPI-DynFusion alternatively applies an attention-guided mechanism that trains on weight contributions from different branches (CNN for local features, Transformer/BiGRU for global patterns) on the grounds of a certain input. The comparative significance of local versus global features differs across different enhancer-promoter pairs, acknowledging this resilience. The approach pulled off an AUROC of 96.2%.

The Transformer architecture is used by EPI-Trans [12] for precise interaction prediction. DNA2Vec embeddings and CNN layers are used by the model for primary feature extraction, then merge enhancer and promoter features in the early stages before augmenting them across the Transformer. How Motifs in enhancers relate to motifs in promoters is specifically modeled by the Multi-Head Attention mechanism due to this initial integration. The model obtained an average AUROC of 95.7%, which confirms that attention mechanisms can capture the long-range dependencies deep-rooted between enhancer-promoter connections.

The main inspiration for our project is KansformerEPI [15]. Basic MLP layers are recognized as a restriction for modelling the complicated nonlinear relationships in gene regulation by the authors. Their solution merges Kolmogorov-Arnold Networks with a Transformer encoder.Rather than fixed activations at nodes,KAN layers apply learnable activation functions on network edges which allows more adaptable approximation of nonlinear functions. The model represented strong performance (AUC greater than 0.90) through many cell lines, with better stability in contrast to MLP-based options.

Efficiency and performance can coexist, as shown by EPInformer [13]. Building on the Enformer architecture, this Transformer-based framework amalgamates sequences, epigenomic signals, and Hi-C contact data to predict gene expression. Prominently, while using only 0.2

EPIPDLF [14] focuses on robustness by adversarial training. The model learns to depend on proper biological signals rather than misleading correlations that might appear from noise in the training data by using Projected Gradient Descent to generate perturbed inputs during training. This approach earned strong benchmark outcomes and advised that robustness may correlate with learning biologically meaningful features.

How data quality affects model performance is highlighted by work on negative set construction [18]. Random negative sampling generates datasets where non-interacting pairs are usually farther faraway than interacting pairs, allowing models to earn high accuracy by directly learning this distance bias. Class-Balanced Maximum-Flow and Class-Balanced Gibbs Sampling methods create the distance distribution of positive examples that match the negative sets. When we train models on these balanced datasets, they exhibit significantly better recall and precision scores when tested carefully.

In general evaluation practices, the LOCO-EPI [19] benchmarking framework [19] exhibits weaknesses. Traditional random train-test divisions can result in data leakage when the same genomic regions appear in both train-test sets. Leave-One-Chromosome-Out evaluation illustrates this by defying whole chromosomes for testing. Under this policy, multiple published models perform near chance levels, which reveals that instead of proper biological learning, reported accuracies occasionally exhibit overfitting. The authors present a multi-branch network that maintains performance even under strict evaluation like LOCO.

The practical value of these methods is displayed by disease-specific implementations. The Transformer-GAN model was used to handle the scarcity of disease-specific interaction data in work on periodontitis [23]. Key inflammatory genes regulated by distal enhancers are specified by the approach, displaying how computational predictions can create practically significant hypotheses about disease mechanisms.

2.2.3 The Interpretability and Feature Reduction Frontier

Deep learning models present ongoing difficulties to be able to make sense of what they have learn. To fulfill this need, multiple approaches have appeared.

The input regions that contribute most to predictions are highlighted in Class Activation Maps present visual explanations. In genomic contexts, this can expose which sequence positions or motifs control the classification of a sequence as an enhancer or the prediction that two elements communicate. Computational mutagenesis experiments, where high-importance regions are disordered and predictions re-evaluated, help authenticate that recognized regions are properly functional.

Reinforcement learning approaches as PPO provide a separate path to interpretability by feature selection. By training an agent to pick which features to incorporate while maximising prediction accuracy, these methods recognize minimal feature sets enough for good performance. The resulting feature importance rankings can guide biological interpretation and experimental follow-up.

Attention weights from Transformer models provide yet another window into model behaviour. High attention between certain sequence positions suggests that the model has learned that those positions relate to each other. However, interpreting attention weights requires caution, as they do not always correspond directly to feature importance in a causal sense.

2.2.4 Related Research and Similar Applications

Several foundational papers established the architectures that current methods build upon.

DanQ [5] pioneered the combination of CNNs and BiLSTMs for genomic sequence analysis. The CNN component captures local motif patterns while the BiLSTM learns how these patterns arrange themselves along the sequence. By demonstrating that sequential dependencies matter for grasping regulatory DNA, this mixed approach surpassed pure-CNN models

DeepSite [7] applied an equivalent hybrid BLSTM-CNN architecture to predicting protein binding sites. Capturing long-range dependencies by recurrent layers remarkably improves sensitivity compared to CNNs alone, as confirmed by the work

DeFine [8] enhanced prediction from binary classification to quantitative regression, instead of predicting just the presence or absence of binding, it predicts the actual intensity of ChIP-seq signals. This change matters because regulatory interactions work on a continuum instead of as simple on-off switches. Prioritisation of disease-associated variants based on their predicted functional impact is enabled due to the ability to predict quantitative changes in binding intensity.

Feedbacks of enhancer-promoter biology present essential context for computational work. With both processes reinforcing each other, studies examining whether chromatin loops cause transcription or vice versa [20] offer a mutual causal relationship. This dynamic interplay indicates that static computational models may overlook important feedback effects.

The mechanisms by which cancers hijack gene regulation is demonstrated by the work on enhancer deregulation in cancer [21]: structural variants that reorder TAD boundaries (enhancer hijacking), extrachromosomal DNA that generates novel regulatory environments, and phase-separated condensates (onco-condensates) that condense transcriptional machinery improperly. The need for models sensitive to chromatin architecture, not just sequence features, is highlighted by these mechanisms.

Studies of the molecular machinery underlying chromatin looping [22] recognise the key players: cohesin for loop extrusion, CTCF for anchoring loops at particular sites, and transcription factors such as LDB1 that can mediate interactions along different mechanisms. A complete computational model would ideally learn to identify signatures of these unlike interaction modes.

2.3 Gap Analysis

2.3.1 Synthesis and Gap Analysis

After reviewing a all the current literature it appears that, Deep learning methods have archives great performance in identification of enhancer and prediction task. Many aspect of problems are addressed by some architectural innovation like attention mechanisms, Transformer encoders and hybrid CNN-RNN designs. Tools have stated to open black box that's help researchers to identify some important features. So far remarkable gap still remain. So many innovation exist separately with sophisticated sequence encoding which is use for enhancer identification but not combined in the prediction pipelines and simplistic input representation in advanced architectures. There are some studies use very strict cross chromosome validation while some other based on random splits which can increase reported performance. Some complexity of gene regulation including phase separation, cooperative binding and dynamic state are still incompletely captured by current models.

2.3.2 Fundamental Gaps

There are three fundamental gaps that motivate us for this project:

Gap 1: Disconnect Encoding-Architecture. POCD-ND is very effective for enhancer identification, capturing density and distribution information. POCD-ND is based on position aware encoding. On the other hands, KansformerEPI and EPI-Trans are the advance prediction models which often use basic embedding or One-Hot encoding as input. A fidelity gap is created by this dis-connectivity : The input representation lacks the positional subtlety that sophisticated architectures can potentially exploit. Feeding POCD-ND encoded sequences into a Kansformer backbone offers an opportunity to synergise these advances.

Gap 2: Inadequate Nonlinear Fusion. KansformerEPI introduces KAN layers to model nonlinearity, but uses relatively simple methods to combine sequence and epigenetic features. EPI-DynFusion demonstrates the value of dynamic, context-dependent fusion, but does not employ the flexible nonlinear modelling of KANs. Integrating KAN-based fusion that can learn complex weighting functions for combining different data modalities represents a natural next step.

Gap 3: Limited Interpretability in Interaction Models. Reinforcement learning-based feature selection has proven powerful for enhancer classification in DeepEnhancerPPO, but has not been applied to the more complex task of interaction prediction. Extending PPO-based feature reduction to identify features relevant for interactions (rather than just for classifying individual elements) would provide new insight into what drives enhancer-promoter communication.

2.3.3 Comparative Summary

Table **??** summarises how existing methods address different aspects of EPI prediction and how our proposed approach aims to combine their strengths.

graphicx

Table 2.1: Unified comparison of existing approaches, identified gaps, and the proposed POCD-KansformerEPI

Dimension	**KansformerEPI**	**TransEPI**	**PDCNN**	**DeepEnhancerPPO**	**EPI-DynFusion**	**Proposed: POCD-KansformerEPI**
Primary Task	EPI Prediction	EPI Prediction	Enhancer Identification	Enhancer Identification	EPI Prediction	**EPI Prediction**
Sequence Encoding	Embeddings	OHE	**POCD-ND**	4-mer Embedding	DNA2Vec	**POCD-ND**
Architecture	**KAN + Transformer**	Transformer	CNN	ResNet + Transformer	Transformer + BiGRU	**KAN + Transformer + BiLSTM**
Nonlinear Modeling	Yes	Limited	Moderate	Moderate	Yes	**Strong (KAN-based)**
Feature Fusion Strategy	Static	Static	N/A	Concatenation	**Dynamic Attention**	**Dynamic + KAN-based Fusion**
Multi-task Learning	Yes	No	No	No	No	**Yes**
Interpretability Mechanism	Basic	None	None	**RL (PPO)**	Attention / CBAM	**RL (PPO) + CAMs**
Cross Cell-line Generalization	Yes	Partial	No	Limited	Yes	**Yes**

2.4 Summary

This chapter has established the biological and computational foundations for our project. Enhancers and promoters communicate across large genomic distances through chromatin looping, creating a prediction problem that requires understanding both sequence features and three-dimensional genome organisation. Experimental data from ChIP-seq, accessibility assays, and chromosome conformation capture techniques provide the raw material for computational models. The literature review traced the evolution of methods from simple classifiers to sophisticated deep learning architectures. Advance models like Transformer attention mechanisms, hybrid CNN-RNN design, position aware sequence encoding and KAN layer use for nonlinear modelling. Class activation Maps and reinforcement learning based feature selection can help to connect biological mechanisms prediction. In our Gap analysis we found three main opportunities: 1.Combining position aware encoding with advance architectures. 2.Fusion implementation feature based on KAN. 3.Extending interpretability methods, for Interaction prediction. The specific contribution of our project is defined by these gaps.

Chapter 3

Project Design

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

Functional requirements:

- Enhancer and promoter supports inputs of raw DNA sequences.
- Epigenetic changes in a cell cycle like histone modifications, accessibility signals processes as an optional feature data .
- From input sequences it creates POCD-ND for encoded representations.
- Predicts enhancer-promoter interaction status in binary.
- Generate an auxiliary output by estimating the genomic distance between input pairs.
- Generate Intuitive outputs like importance ranking and activation maps.
- Large scale genomic analysis compatibility with batch processing.

Non-functional requirements:

- **Performance:** On cross-cell-line validation achived above 90 percent AUROC
- **Efficiency:** Reasonable time queries processing for practical use
- **Generalisability:** Maintain good performance on real world unseen data and handles well
- **Interpretability:** Besides raw prediction it also provide biological insights

3.1.2 Context Diagram

The system also acts on external entities:

- **Genomic Databases:** BENGI, ENCODE, CATlas and Enhancer Atlas.
- **User Interface:** It supports queries of specific genomic sequences.

- **Output Consumers:** Shows the analysis pipelines and companion as a visualization tool and also provides experimental validation.

From the genomic databases to the model's predictions, the system provides interpretability outputs to the users.

3.1.3 Data Flow Diagram Level 1

The system processes data through the following stages:

1. **Data Ingestion:** Raw sequences and coordinates enter the system
2. **Preprocessing:** Sequences are validated, standardised, and prepared for encoding
3. **POCD-ND Encoding:** Sequences transform into position-aware density matrices
4. **Epigenetic Processing:** Chromatin features are extracted and normalised
5. **Feature Fusion:** Sequence and epigenetic streams merge in the Kansformer
6. **Prediction:** Classification and regression heads generate outputs
7. **Interpretation:** PPO agent and CAM modules produce explanatory outputs
8. **Output Formatting:** Results package for delivery to users

3.1.4 UI Design

The system will provide:

- Command-line interface for batch processing and pipeline integration
- Configuration files for specifying model parameters and data paths
- Output files in standard formats (TSV for predictions, image formats for visualisations)
- Logging of training progress and evaluation metrics

3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.

3.2.1 Data Acquisition and Preprocessing

Training data comes from the BENGI repository, which provides validated enhancer-promoter pairs from multiple cell types. Positive examples are pairs where experimental evidence (Hi-C, ChIA-PET) supports physical interaction. Negative examples use the distance-matched strategy: for each positive pair at distance d, we sample a non-interacting pair at approximately the same distance. This prevents the model from learning distance as a shortcut.

Epigenetic features extract from a 2.5 Mbp window centred on each pair:

1. CTCF binding (loop anchors)
2. DNase accessibility (open chromatin)
3. DNA methylation (often inversely correlated with activity)
4. H3K4me1 (enhancer priming)
5. H3K4me3 (promoter activity)
6. H3K27me3 (Polycomb repression)
7. H3K36me3 (transcription elongation)
8. H3K9me3 (heterochromatin)

Signals are binned into 500 bp segments and averaged within each bin, creating a context matrix $C \in \mathbb{R}^{\text{Bins}\times 8}$.

3.2.2 POCD-ND Encoding

The encoding proceeds in four steps:

Step 1: K-mer Segmentation. Input sequences of length L split into overlapping trinucleotides (k=3), based on prior optimisation showing this balances feature richness with manageable sparsity.

Step 2: Position-Specific Frequency Matrices. Two matrices A^{pos} and A^{neg} of size $64 \times (L-2)$ count how often each trinucleotide appears at each position in positive and negative sequences respectively.

Step 3: Density Normalisation. Frequencies convert to densities by dividing by total k-mer counts:

$$A^{pos}_{den} = \frac{A^{pos}}{NS^{pos}}, \quad A^{neg}_{den} = \frac{A^{neg}}{NS^{neg}}$$

Step 4: POCD-ND Calculation. The final encoding combines propensity (ratio of densities) with weighting by minimum density to avoid artifacts from rare k-mers:

$$PSTNPss_{i,j} = \frac{A^{pos}_{den}(i,j)}{A^{neg}_{den}(i,j)}$$

$$POCD\text{-}ND_{i,j} = PSTNPss_{i,j} \times \min\big(A^{pos}_{den}(i,j), A^{neg}_{den}(i,j)\big)$$

3.2.3 Neural Architecture

Sequence Branch: The POCD-ND matrix feeds into 1D CNN layers that detect local motif patterns. A BiLSTM layer then processes the CNN output bidirectionally, capturing sequential dependencies and the "grammar" of motif arrangements.

Epigenetic Branch: The context matrix C passes through CNN layers with max pooling to extract salient chromatin features while compressing dimensionality.

Kansformer Encoder: Outputs from both branches concatenate to form fused features Z. The encoder applies:

1. Multi-Head Self-Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

2. KAN Layers replacing standard feed-forward networks:

$$Z_{k+1,q} = \sum_{p=1}^{n_k} \varphi_{k,q,p}(Z_{k,p})$$

   where φ functions are learnable B-spline activations on network edges.

Prediction Heads:
Classification head:

$$p = \sigma(KAN_1(H))$$

Distance regression head:

$$d_p = KAN_2(H)$$

Loss Function:

$$\mathcal{L} = \mathcal{L}_{class}(y, p) + \lambda\,\mathcal{L}_{dist}(d_p, d_t)$$

where $\mathcal{L}_{class}$ is Binary Cross-Entropy, $\mathcal{L}_{dist}$ is Mean Squared Error, and λ weights the auxiliary task.

3.2.4 Interpretability Modules

PPO Feature Selection: A reinforcement learning agent receives the fused feature vector as state, outputs a binary mask selecting which features to retain, and receives classification accuracy as reward. Over training, the agent learns to identify features essential for prediction.

Class Activation Maps: CAMs visualise which input positions contribute most strongly to predictions. We validate identified regions through in silico mutagenesis, scrambling high-importance positions and confirming that predictions change appropriately.

3.3 Project Plan

The project proceeds through the following phases:

1. **Weeks 1-2:** Data collection and preprocessing pipeline development
2. **Weeks 3-4:** POCD-ND encoding implementation and validation
3. **Weeks 5-7:** Neural architecture implementation (sequence branch, epigenetic branch, Kansformer encoder)
4. **Weeks 8-9:** Multi-task learning framework and loss function tuning

5. **Weeks 10-11:** Interpretability module integration (PPO, CAMs)
6. **Weeks 12-13:** Cross-cell-line validation experiments
7. **Week 14:** Ablation studies and performance analysis
8. **Weeks 15-16:** Documentation and final report preparation

3.4 Task Allocation

Project tasks distribute across team members as follows:

- **Data Engineering:** Database integration, preprocessing pipelines, negative sampling implementation
- **Encoding Development:** POCD-ND implementation, validation against published results
- **Architecture Implementation:** CNN, BiLSTM, Transformer, and KAN layer coding
- **Training Infrastructure:** Loss functions, optimizers, learning rate schedules, checkpointing
- **Interpretability:** PPO agent implementation, CAM visualisation tools
- **Evaluation:** Benchmark comparisons, ablation studies, statistical analysis
- **Documentation:** Code documentation, user guides, final report

3.5 Summary

This chapter has presented the complete design for the POCD-KansformerEPI system. Requirements analysis established what the system must accomplish and the constraints it must satisfy. The detailed methodology walked through each component from data preprocessing through POCD-ND encoding to the hybrid Kansformer architecture and interpretability modules. The project plan and task allocation provide a roadmap for implementation.

The design addresses the three gaps identified in the background chapter: it combines position-aware POCD-ND encoding with the advanced Kansformer architecture, implements KAN-based feature fusion, and extends interpretability through PPO-based feature selection and CAM visualisation. Successful implementation should yield a model that not only achieves strong prediction performance but also provides biological insights into the sequence and chromatin features that govern enhancer-promoter communication.

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

This chapter outlines the technical and operational standards governing the POCD-KansformerEPI framework for enhancer-promoter interaction (EPI) prediction, along with design constraints, cost analysis, and characterisation as a complex engineering problem.

5.1 Compliance with the Standards

Implementation of a combination of different frameworks for enhancer-promoter interaction (EPI) prediction requires obedience to multiple technical and operational standards of software, hardware, and communication sectors. This section highlights the compliance mechanisms predicted for the proposed system.

5.1.1 Software Standards

The software architecture integrates multiple important standards to assure quality and reproducibility.

Programming and Development Standards: Code implementation follows Python Enhancement Proposal (PEP) 8 standards to ensure consistency and maintainability.Open source frameworks like TensorFlow or PyTorch with GitHub repositories as version controller are being utilised for developing models. To keep a track of the performance metrics, hyperparameter configurations, and other runtime diagnostics, all the training models include logging. [1, 15].

Alternatives Considered:

- **MATLAB:** Though it offers robust numerical computing, it lacks the large-scale deep learning environment and the significant community support that is available for Python.
- **Julia:** It offers great computational speed but for genomics applications, it has a smaller environment. Also it provides fewer pre-trained models than Python.
- **R:** Great for statistical analysis but for deep learning framework the support is very limited compared to Python.

Rationale: Python with PyTorch/TensorFlow is selected for its vast bioinformatics libraries (Biopython, pybedtools), optimized deep learning frameworks, and a common choice in computational biology research ensuring reproducibility.

Machine Learning and Data Processing Standards: We preprocess the Dataset following the ENCODE and GEO standards for genomic data representation [24]. POCD-ND encoding technique is used for DNA sequence encoding to have consistent position-aware representation that ensures compatibility with transformer and CNN modules. Data is augmented using the sliding window approach (window size k=3, step size s=1) as validated in sequence-based EPI prediction literature. [12].

Benchmarking and Validation Standards: Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision-Recall Curve (AUPR) [18, 19] are widely accepted evaluation metrics to satisfy the bioinformatics standards. The Leave-One Chromosome-Out (LOCO) strategy is used in Cross-validation to tackle sequence homology-based data leakage[19]. We followed class-balanced negative sampling protocols with a 1:20 positive-to-negative ratio across six ENCODE cell lines (GM12878, HeLa-S3, HUVEC, IMR90, K562, NHEK) [11, 12] to train the models.

5.1.2 Hardware Standards

Strict hardware requirements must be met by the computing infrastructure that facilitates model training and inference

GPU and Computing Resources: The minimum GPU needed to train a model is an NVIDIA Tesla K40C or any other GPU equivalent of 12 GB VRAM. The recommended hardware for training multiple models as part of a model ensemble is an NVIDIA GeForce RTX 2080 Ti (11GB VRAM) with 256GB of system memory. For validating the model ensemble results, this configuration will also be used. Training of each model takes approximately 33.2-35.0 minutes for each cell line on recommended hardware [11].

Alternatives Considered:

- **Cloud Computing (AWS/GCP):** While scalable, cloud computing solutions incur ongoing expenses and have high data transfer costs.
- **CPU-only Training:** Removes need to train on GPUs, however, increases training times by an order of magnitude.
- **TPU (Tensor Processing Units):** Optimized for TensorFlow-based architectures and performance, TPUs are not flexible enough to be used as a substitute for custom KAN Layer Implementations.

Rationale: Because local GPU hardware is the most cost-effective option when dealing with large training run times, while providing the ability to maintain user data in house and allowing for greater flexibility in implementation of custom KAN Layers, local GPU hardware has been chosen as the preferred architecture.

Memory and Storage Specifications: A total of 4–5GB of memory will be required as a base memory footprint for the hybrid POCD-KansformerEPI architecture[11], while a minimum of 50–100 GB of local disk space will be required for storing the augmented datasets and checkpoints of the models. The system will require at least 8 GB of system RAM, however, 16 GB will be preferred to ensure smooth multi-tasking during the development phase of this project.

Computational Specifications: It will require multi-Core CPU (4–8 cores) Ubuntu 18.04 LTS or equivalent Operating System, Python 3.7 or higher and faramwork like TensorFlow 2.x or PyTorch 1.x.

5.1.3 Communication Standards

Data Integration and Interoperability: The exchange of genomic information occurs in the form of FASTA format for DNA sequences; FASTQ format is used when quality-annotated sequence information is exchanged. Epigenomic annotation formats for ChIP-seq, ATAC-seq and DNA methylation follow the ENCODE data standards regarding peak calling and signal representation [24]. Model Output Formatting: The model output is formatted as a comma-separated values file (CSV) with structural metadata including cell-line identifiers, EPI probability scores and confidence intervals.

Documentation and Reporting Standards: Technical documentation complies with IEEE Standard 1016-2009 for software design specifications. Model interpretability reports follow explainable AI (XAI) standards, including saliency maps and attention weight visualisation [25]. Performance reporting adheres to FAIR (Findable, Accessible, Interoperable, Reusable) data principles and bioinformatics standards.

Version Control and Reproducibility: Git repositories with detailed commit message documents all code changes and experimental progression. Experimental tracking of Jupyter Notebooks uses Markdown for the explanation. Docker containerization allows for consistent and reproducible results on any given computing environment.

5.2 Design Constraints

Only mention the constraints that are related to the design of your project. This list is not complete.

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

The enhancer-promoter interaction prediction framework is designed to minimise financial overhead while maintaining high research quality.

Primary Budget:

Item	Estimated Cost (USD)
Software Licenses (Open-source)	$0
GPU Hardware (NVIDIA RTX 2080 Ti)	$700–1,000
Storage (1TB SSD)	$100
Development Workstation	$1,500–2,000
Cloud Backup (Annual)	$100
Total	$2,400–3,200

Alternative Budget (Cloud-Based):

Item	Estimated Cost (USD)
AWS/GCP GPU Instance (500 hours @ $3/hr)	$1,500
Cloud Storage (100GB for 12 months)	$250
Development Laptop (Mid-range)	$800
Total	$2,550

Rationale: The primary budget with local GPU hardware was selected because: (1) it provides unlimited training iterations without per-hour costs, (2) faster iteration during development without network latency, (3) complete control over software environment, and (4) hardware remains useful for future projects after completion.

Revenue Model: As an academic research project, direct revenue generation is not the primary objective. However, potential value creation includes:

- Publication in peer-reviewed bioinformatics journals.
- Open-source tool adoption by the computational biology community.
- Foundation for future research grants in gene regulation prediction.
- Potential licensing of novel architectural components for commercial applications.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

A complex engineering-problem spanning multiple dimensions of computational biology, integration of genomic data and architecture design of machine learning is represent by enhancer-promoter interaction (EPI) prediction challenge. Table 5.1 provides a extensive mapping of the 7 complex parameters to our research domain.

P1: Depth of Knowledge Required

Very carefull Integration of multiple knowledge domains is required for this research. For understanding enhancer-promoter regulatory interaction it requires proficiency knowledge of:

- **Molecular Biology:** Molecular biology is study of chromatin organisation mechanism, factor binding, long range DNA interaction and transcription factor binding [20, 22].

Table 5.1: Mapping with complex problem solving.

P1 Depth of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Interdependence
√ Satisfied	√ Satisfied	√ Satisfied	◦ Partial	√ Satisfied	√ Satisfied	√ Satisfied

- **Computational Genomics:** this is the study of Processed sequencing data from ChIA-PET, Hi-C, and ChIP-seq experiments [24].
- **Deep Learning Architectures:** used for Understand chromatin looping and topologically associating domains (TADs)[5, 15].
- **3D Genome Topology:** used for Understand chromatin looping and topologically associating domains (TADs) [20].

Knowledge Profile Mapping:

Knowledge Domain	**Level Required**	**Rationale**
Molecular Biology	Advanced	Understanding gene regulation mechanisms
Deep Learning	Expert	Designing hybrid KAN-Transformer architectures
Bioinformatics	Advanced	Genomic data processing and validation
Mathematics	Intermediate	Loss function design, attention mechanisms
Software Engineering	Intermediate	Reproducible pipeline development

P2: Range of Conflicting Requirements

Competing intention that avert simple optimization is the fundamental characteristic of the e EPI prediction problem. [11, 12]:

- **Accuracy vs. Efficiency:** It is important to maintain the balance between maximising model accuracy (AUROC/AUPR) and computational efficiency constraints(training time 40 minutes, memory 5 GB)
- **Generalisation vs. Specificity:** In individual cell line Cell-type-specific model achieve superior accuracy but in generalise tissue it's fail
- **Sequence-only vs. Multi-omics:** Reproducibility increase if it relying on DNA sequence but it also ignore epigenomic context crucial when in terms of biological fidelity
- **Interpretability vs. Complexity:** IIncreasing interpretability of a model causes computational overhead.

P3: Depth of Analysis

To solve the EPI prediction problem it required systematic analysis through computational and biological scales. [1, 12]:

- **Sequence Level:** positional nucleotide encoding using POCD-ND,DNA k-mer composition and discovery of transcription factor motif
- **Architecture Level:** Kansformer configuration and characteristics optimisation is filter by CNN (embedding dimensions, KAN layer depth, attention heads).
- **Genome Level:** topologically associating domains (TADs) which constrain permissible long-range interactions analysis must accounted
- **Cell-type Level:** SyDoing systematic comparison in 6 ENCODE cell line finds EPI patterns vary substantially across tissues

P4: Familiarity of Issues (Partially Satisfied)

In EPI prediction, Integration of POCD-ND encoding with Hybrid KAN-Transformer architecture become novel research territory After the deep learning has become well-established in genomics in prominence publication like DeepSite [7] and DanQ [5]. Previous work has established transformer-based approaches and CNN separately. [11, 15].

P5: Extent of Applicable Codes

ThA well-defined ecosystem of bioinformatics standards is required for this research. [19, 24]:

- compatibility with publicly available experimental resources ensure by ENCODE data standards
- it is need to maintain principles like FAIR (Findable, Accessible, Interoperable, Reusable) for structure data stewardship and code availability
- Rather than raw accuracy Bioinformatics evaluation standards mandate use of AUROC and AUPR metrics
- sequence homology from inflating generalisation estimates by LOCO (Leave-One-Chromosome-Out) cross-validation paradigm prevents [19].

P6: Extent of Stakeholder Involvement

Engagement of multiple, sometimes conflicting stakeholder groups is happened in the EPI prediction research [21]:

- **Computational Biologists:** Mechanistic understanding and Prioritise interpretability
- **Genomics Researchers:** Proper benchmarking on rigidity and dataset diversity.
- **Clinical Researchers:** For translational applications it need to seek disease-specific predictions.
- **Funding Organisations:** reproducibility and resource efficiency is required

P7: Interdependence of Components

significant component interdependencies is exhibits by The POCD-KansformerEPI framework [11, 12]:

- CNN feature extraction affects downstream by POCD-ND encoding quality directly.
- How efficiently the Kansformer learns long-range dependencies is affected by Position-aware nucleotide encoding
- coupling between prediction heads by Multi-task learning (classification + distance regression)
- Feature selection agent training of PPO is depends on stable base model predictions.

5.4.2 Engineering Activities

the project to complex engineering activity characteristics. Is mapped by the Table 5.2

Table 5.2: Mapping using complex engineering activities and other.

A1 Range of Resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for Society and Environment	A5 Familiarity
√ Satisfied	√ Satisfied	√ Satisfied	√ Satisfied	○ Partial

A1: Range of Resources

The project requires diverse resources from different categories:

- **Computational:** Data: BENGI benchmarked data, ENCODE datasets multiple cell line epige nomic profiles.
- **Data:** Computational: storage systems, cloud backup infrastructure and GPU clusters.
- **Software:** Human: machine learning, expertise in molecular biologyand software engineering knowledge.
- **Human:**Software: Bioinformatics tools with Deep learning frameworks and visualisation libraries

A2: Level of Interaction

Complex interactions is involved in the system:

- **Component Interactions:** POCD-ND encoder → CNN → BiLSTM → Kansformer → Prediction heads.

- **Data Flow Interactions:** Epigenetic streams and Sequence merge using dynamic Mechanisms
- **Stakeholder Interactions:** Understandin between the computational and the experimental biology perspectives is very much needed.

A3: Innovation

Several innovative contributions is introduced by the project:

- First integration of Kansformer architecture with POCD-ND encoding for EPI prediction.
- Dynamic feature fusion mechanism(Novel KAN-based) combining sequence and epigenomic data.
- PPO-based feature is extend for selection from enhancer classification to interaction prediction.
- Multi-task learning framework combining binary classification with distance regression.

A4: Consequences for Society and Environment

The research has potential some societal impacts like:

- **Positive:** Its Improves understanding of potential applications with gene regulation mechanisms in identifying disease-associated based on genetic variants with using advancement of computational biology tools[21].
- **Environmental:** OOptimised model are designing for minimises computational energy consumption.
- **Healthcare:**Using dysregulated gene expression Predictions may eventually contribute to understanding diseases , including cancer and other developmental disorders

A5: Familiarity (Partially Satisfied)

The POCD-KansformerEPI architecture represents novel territory While individual components (CNNs, Transformers, genomic data processing) are well-established and use their specific combination .The integration of KAN-based nonlinear modelling with POCD-ND encoding has not been previously explored for enhancer-promoter interaction prediction[1, 15].

5.5 Summary

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

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